KERNEL DIMENSION OF THE $\mathbb{Z}/2$ CONNECTION LAPLACIAN ON A FINITE SIMPLE GRAPH
(A FORMALISATION IN LEAN 4 / MATHLIB V4.11.0)

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Abstract. We present a fully formalised, machine-checked proof library in Lean 4 / Mathlib v4.11.0 for the kernel-dimension theory of the $\mathbb{Z}/2$ *connection Laplacian* of a finite simple graph G. A connection graph is a finite simple graph equipped with a designated set $W \subseteq E(G)$ of *wrap edges*; on each edge e we attach a 2×2 real matrix (the Pauli σ_x on wrap edges, $\mathbb{I}_2$ otherwise), yielding a block Laplacian L_{Mob}. Our principal results, each proved from the three standard Lean axioms (`propext`, `Classical.choice`, `Quot.sound`) and nothing else, are the following.

(i) A Hadamard–Kronecker conjugation decomposes L_{Mob} into the block-diagonal sum of the ordinary (*scalar*) Laplacian and a *signed* Laplacian, both on G.

(ii) The orientation double cover $\widetilde{G}$ of G inherits a trivial connection, and its scalar Laplacian splits similarly.

(iii) A base connected component $C \in \pi_0(G)$ is *balanced* iff the induced map $\pi_0(\widetilde{G}) \to \pi_0(G)$ has fibre of cardinality 2 over C.

(iv) The kernel dimension of the signed Laplacian equals the number of balanced components; the full $\mathbb{Z}/2$-connection Laplacian has kernel dimension $\#\pi_0(G)+\#\{\text{balanced components}\}$ (Möbius) and $2\,\#\pi_0(G)$ (flat).

(v) On the n-cycle the Fourier basis diagonalises both modes explicitly.

The bespoke arguments are the Hadamard rotation of part (i)–(ii) (a Kronecker-conjugation computation) and the double-cover lift of part (iii) (a walk induction). Along the way we observe and correct an incorrect statement that appeared in our pre-registered plan: the identity $\dim\ker L^{\mathrm{sg}}(G) = \#\pi_0(G) - \#\{\text{components of odd total wrap}\}$ holds only on $\mathbb{Z}/2$-cycles, and must in general be replaced by the cohomological formula $\dim\ker L^{\mathrm{sg}}(G) = \#\{\text{balanced components}\}$, which is the content of Theorem 6.2. The Lean sources and a numerical cross-check on every simple graph of order ≤ 5 (3,456 configurations, zero disagreements) – and an extended fuzzer sweep at orders $n \leq 7$ (456,950 configurations, also zero disagreements) accompany this paper.

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1. Introduction

Let $G = (V, E)$ be a finite simple graph and let $W \subseteq E$ be a distinguished set of *wrap* edges. Attach to each $e \in E$ a 2×2 real matrix $\Phi(e)$ by

$$\Phi(e) := \begin{cases} \sigma_x & \text{if } e \in W, \\ \mathbb{I}_2 & \text{if } e \notin W, \end{cases} \qquad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

The $\mathbb{Z}/2$-*connection Laplacian* of (G, W) is the $V \times V$ block matrix with 2×2 blocks

$$L_{\mathrm{Mob}}[u, v] := \begin{cases} \deg(u)\, \mathbb{I}_2 & u = v, \\ -\Phi(\{u, v\}) & \{u, v\} \in E, \\ 0 & \text{otherwise.} \end{cases} \tag{1}$$

Equivalently, $L_{\mathrm{Mob}} = D \otimes \mathbb{I}_2 - \sum_{e \in E} \mathbf{1}_e \otimes \Phi(e)$, where $\mathbf{1}_e$ is the $V \times V$ edge-indicator of e (both Kronecker factors ordered vertex-major, matching the $V \times \mathrm{Fin}\, 2$ index set of Equation (1)). The *flat* Laplacian L_{flat} is obtained by the same recipe with $\Phi \equiv \mathbb{I}_2$, i.e. by ignoring the wrap structure; it agrees with two copies of the ordinary graph Laplacian. The guiding question answered in this paper is:

In terms of (G, W)*, what is* $\dim \ker L_{\mathrm{Mob}}$ *and when is it strictly smaller than* $\dim \ker L_{\mathrm{flat}}$*?*

This is the finite-dimensional, discrete analogue of the spectrum of the Laplacian on a flat $\mathbb{Z}/2$ line bundle over a graph: the kernel detects global sections of the bundle, and the flat bundle always has two independent sections per component. The Möbius bundle admits a non-vanishing section over a component exactly when the *wrap* holonomy of the bundle around every cycle in that component is trivial; this is the condition of *balance* in the language of Harary, Cartwright and Zaslavsky [1, 2].

Organisation and results. Section 2 sets the notation and fixes the Lean-formalised data. Section 3 proves the Hadamard–Kronecker decomposition that reduces the block Laplacian to two ordinary Laplacians (one scalar, one signed). Section 4 constructs the orientation double cover and proves that its scalar Laplacian splits as the direct sum of the scalar and signed Laplacians of G. Section 5 formalises the balance notion as the vanishing of a $\mathbb{Z}/2$-cohomology class and computes the fibre cardinality of the induced map on π_0. Section 6 assembles everything into the *recognition theorems* which count kernel dimension by balanced components. Section 7 specialises to the n-cycle, where the Fourier basis diagonalises both Laplacians. Section 8 summarises the formal Lean development.

The main structural decomposition, cover-splitting, cohomological recognition, kernel-dimension, and auxiliary identities are machine-checked in Lean 4 against Mathlib v4.11.0. For the cycle spectra, the Lean development proves the per-mode eigenvalue identities; the displayed spectrum formulas are the corresponding mathematical assembly (see Remark 7.3 in Section 7). The complete proof scripts and an exhaustive numerical verification are included in the accompanying Lean project.

2. Preliminaries

2.1. **Connection graphs.** A *connection graph* is a quadruple $G = (V, E, \mathrm{Adj}, W)$ where V is a finite type, $\mathrm{Adj} \subseteq V \times V$ defines a (mathematically simple) graph on V, $E = \{\{u, v\} : \mathrm{Adj}(u, v)\}$ is its edge set and $W \subseteq E$ is an arbitrary subset, the *wrap* edges. In Lean we keep the wrap data as a predicate `wrap : G.graph.edgeSet → Prop` and carry explicit `Fintype`/`DecidableEq`/`DecidableRel` instances throughout.

2.2. **Pauli and Hadamard.** The 2×2 Pauli matrix $\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ is involutive, symmetric and has eigenvalues ± 1, with symmetric eigenvector $(1, 1)^\top$ and antisymmetric $(1, -1)^\top$. The *Hadamard rotation*

$$\widehat{R} := \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \widehat{R}\,\widehat{R} = 2\,\mathbb{I}_2, \tag{2}$$

simultaneously diagonalises σ_x and $\mathbb{I}_2$ in the sense that $\widehat{R}\,\sigma_x\,\widehat{R} = 2\,\mathrm{diag}(1, -1)$ and $\widehat{R}\,\mathbb{I}_2\,\widehat{R} = 2\,\mathbb{I}_2$. In the Lean formalisation we keep two copies of $\widehat{R}$: one indexed by $\{0, 1\} = \mathrm{Fin}\,2$ (for the connection-block decomposition of Section 3) and one indexed by Bool (for the cover-sheet decomposition of Section 4). They are the same matrix under the obvious bijection $\mathrm{Bool} \cong \mathrm{Fin}\,2$.

2.3. **Scalar and signed Laplacians.** Let G be a connection graph. The ordinary *scalar* Laplacian $L^{\mathrm{sc}} \in \mathbb{R}^{V \times V}$ is

$$L^{\mathrm{sc}}[u, v] := \begin{cases} \deg(u) & u = v, \\ -1 & \{u, v\} \in E, \\ 0 & \text{otherwise}, \end{cases}$$

i.e. Mathlib's `SimpleGraph.lapMatrix`. The *signed Möbius* Laplacian $L^{\mathrm{sg}} \in \mathbb{R}^{V \times V}$ encodes the wrap sign on the off-diagonal:

$$L^{\mathrm{sg}}[u, v] := \begin{cases} \deg(u) & u = v, \\ +1 & \{u, v\} \in E,\ \{u, v\} \in W, \\ -1 & \{u, v\} \in E,\ \{u, v\} \notin W, \\ 0 & \text{otherwise}. \end{cases}$$

Both are real symmetric and positive semidefinite as quadratic forms on $\mathbb{R}^V$. For L^{sc} this is Mathlib's `SimpleGraph.posSemidef_lapMatrix`. For L^{sg} we use the direct sum-of-squares expression

$$\langle x, L^{\mathrm{sg}} x \rangle = \sum_{\{u,v\} \in E \setminus W} (x_u - x_v)^2 + \sum_{\{u,v\} \in W} (x_u + x_v)^2,$$

which Lemma 9.5 (§9) turns into the Lean fact `signedLaplacianMöbius_posSemidef` (`L13_PSD.lean`). Equivalently — but through a heavier route not used by the Lean proof — PSD follows from the cover splitting (Theorem 4.4): the orientation double cover's scalar Laplacian is PSD, and similarity to $\mathrm{diag}(L^{\mathrm{sc}}, L^{\mathrm{sg}})$ preserves the PSD property of each diagonal block. The Fourier diagonalisation in §7 below regards both matrices as acting on $\mathbb{C}^V$, but the matrices themselves remain real; complexification is a computational convenience, not a domain change.

3. The Block Decomposition

The connection Laplacian L_{Mob} of (1) lives on the product index set $V \times \mathrm{Fin}\,2$. Reindexing via the canonical bijection

$$\varphi : V \times \mathrm{Fin}\,2 \xrightarrow{\sim} V \sqcup V, \qquad (u, 0) \mapsto \iota_\ell u,\ (u, 1) \mapsto \iota_r u,$$

we get a matrix on the sum index $V \sqcup V$; the block-diagonal decomposition then becomes a genuine *block-diagonal* form `Matrix.fromBlocks`.

Theorem 3.1. *Let G be a connection graph and $\epsilon \in \{\mathit{flat}, \mathit{Mob}\}$. There exists an invertible $P \in \mathbb{R}^{(V\times\mathrm{Fin}\,2)\times(V\times\mathrm{Fin}\,2)}$ such that*

$$\varphi_*\big(P\,L_\epsilon\,P^{-1}\big) = \begin{pmatrix} L^{\mathrm{sc}} & 0 \\ 0 & B_\epsilon \end{pmatrix}, \qquad B_\epsilon = \begin{cases} L^{\mathrm{sg}} & \epsilon = \mathit{Mob}, \\ L^{\mathrm{sc}} & \epsilon = \mathit{flat}. \end{cases}$$

Proof. Set $P = \mathbb{I}_V \otimes_K \widehat{R}$ (Kronecker product with the Hadamard $\widehat{R}$ of (2) acting on the $\mathrm{Fin}\,2$ fibre). By functoriality of the Kronecker product,

$$P \cdot P = (\mathbb{I}_V \cdot \mathbb{I}_V) \otimes_K (\widehat{R}\,\widehat{R}) = \mathbb{I}_V \otimes_K (2\mathbb{I}_2) = 2\,\mathbb{I}_{V\times\mathrm{Fin}\,2},$$

so $P^{-1} = \frac{1}{2}P$, and in particular $\det P \neq 0$.

A direct entrywise calculation using $\widehat{R}\,\sigma_x\,\widehat{R} = 2\,\mathrm{diag}(1,-1)$ and $\widehat{R}\,\mathbb{I}_2\,\widehat{R} = 2\,\mathbb{I}_2$ yields, for $u, v \in V$ and $i, j \in \mathrm{Fin}\,2$,

$$\tfrac{1}{2}\,\big(P\,L_\epsilon\,P\big)\big[(u,i),(v,j)\big] = \begin{cases} L^{\mathrm{sc}}[u,v] & i = j = 0, \\ B_\epsilon[u,v] & i = j = 1, \\ 0 & i \neq j. \end{cases}$$

Reindexing via φ and using $P^{-1} = \frac{1}{2}P$ turns the left-hand side into $\varphi_*\big(P\,L_\epsilon\,P^{-1}\big)$, giving the claim. The flat case is the special case $W = \emptyset$ of the Möbius case, or equivalently the fact that $\widehat{R}\,\mathbb{I}_2\,\widehat{R} = 2\mathbb{I}_2$ collapses both diagonal blocks to the scalar Laplacian. □

Remark 3.2. In the Lean formalisation `laplacian_decomposes` the two ϵ-branches are proved separately: the Möbius branch uses $P = \mathbb{I}_V \otimes_K \widehat{R}$ as above, while the flat branch takes the simpler $P = \mathbb{I}_{V\times\mathrm{Fin}\,2}$ (identity similarity) and reduces the claim to $L_{\mathrm{flat}} = \mathrm{diag}(L^{\mathrm{sc}}, L^{\mathrm{sc}})$ after reindex via φ. Mathematically the flat case is the specialisation of the Möbius case at $W = \emptyset$, but the two Lean branches are syntactically disjoint.

Remark 3.3. In the Lean script the bijection φ is `prodFinTwoEquivSum` and the Kronecker-product identity is the lemma `mul_kronecker_mul` from `Mathlib.Data.Matrix.Kronecker`. The entrywise calculation is made into a single-sum `Finset.sum_eq_single` reduction that kills V-off-diagonal terms using the $\mathbb{I}_V$ factor of P, and then reduces the remaining $\mathrm{Fin}\,2 \times \mathrm{Fin}\,2$ block to one of three cases by matching on equality of vertices and wrap status.

4. The Orientation Double Cover

The preceding decomposition separates the flat part of G from its wrap data inside a single graph. The cover perspective lifts the wrap data into a *branching* structure on a new, connection-free graph.

Definition 4.1. The *orientation double cover* of the connection graph (G, W) is the simple graph $\widetilde{G}$ on vertex set $\widetilde{V} := V \times \mathrm{Bool}$, with adjacency

$$(u,b) \sim_{\widetilde{G}} (v,c) \iff \{u,v\} \in E \text{ and } \big(\{u,v\} \in W \Leftrightarrow b \neq c\big).$$

We view $\widetilde{G}$ as a connection graph with $\widetilde{W} = \emptyset$: every edge of $\widetilde{G}$ is "interior", the cover has *trivial* connection.

Equivalently, $\widetilde{G}$ is the $\mathbb{Z}/2$-covering space determined by the 1-cochain $\sigma_W : E \to \mathbb{Z}/2$, $e \mapsto \mathbf{1}_{e\in W}$ (which is automatically a 1-cocycle since the graph has no 2-cells; see §5). The non-trivial *deck transformation* $\delta : \widetilde{V} \to \widetilde{V}$, $(v,b) \mapsto (v, \neg b)$ is a free graph involution.

Lemma 4.2. *δ is a graph automorphism of $\widetilde{G}$ of order 2 with no fixed points.*

Proof. Involutivity is immediate from $\neg\neg b = b$. The adjacency $(\delta x) \sim_{\widetilde{G}} (\delta y) \Leftrightarrow x \sim_{\widetilde{G}} y$ follows from the identity $(\neg b) \neq (\neg c) \Leftrightarrow b \neq c$ and fixity of the underlying edge $\{u, v\}$. Absence of fixed points is trivial: $(v, b) \neq (v, \neg b)$. □ □

Lemma 4.3. *For every* $(u, b) \in \widetilde{V}$, $\deg_{\widetilde{G}}(u, b) = \deg_G(u)$.

Proof. We construct a bijection $\mathcal{N}_G(u) \to \mathcal{N}_{\widetilde{G}}(u, b)$ by

$$v \longmapsto \big(v,\ b \oplus \mathbf{1}_{\{u,v\}\in W}\big),$$

where $\oplus$ is the Bool XOR. Correctness is the definition of $\sim_{\widetilde{G}}$; injectivity follows by reading the first coordinate; surjectivity uses the biconditional $\{u, v\} \in W \Leftrightarrow b \neq c$ to recover the unique v with $(u, b) \sim_{\widetilde{G}} (v, c)$. □

We write $\widetilde{L} := L^{\mathrm{sc}}(\widetilde{G})$ for the scalar Laplacian of the cover.

The Hadamard rotation on the Bool**-fibre.** Replace the Fin 2-fibre of Section 3 by the Bool-fibre, viewing

$$\widehat{R} : \mathbb{R}^{\mathrm{Bool}\,\times\,\mathrm{Bool}}, \qquad \widehat{R}[i, j] = \begin{cases} -1 & i = j = \top, \\ 1 & \text{otherwise}, \end{cases}$$

so $\widehat{R}\,\widehat{R} = 2\,\mathbb{I}$ on $\mathbb{R}^{\mathrm{Bool}\,\times\,\mathrm{Bool}}$. Set

$$P_{\mathrm{cov}} := \mathbb{I}_V \otimes_K \widehat{R} \quad \in \mathbb{R}^{\widetilde{V}\times\widetilde{V}}.$$

Theorem 4.4. *Under the canonical bijection* $\varphi : \widetilde{V} \xrightarrow{\sim} V \sqcup V$, $(v, \bot) \mapsto \iota_\ell v$, $(v, \top) \mapsto \iota_r v$, *we have*

$$\varphi_*\big(P_{\mathrm{cov}}\,\widetilde{L}\,P_{\mathrm{cov}}^{-1}\big) = \begin{pmatrix} L^{\mathrm{sc}}(G) & 0 \\ 0 & L^{\mathrm{sg}}(G) \end{pmatrix}.$$

In particular, P_{cov} *is invertible and* $\widetilde{L} \sim L^{\mathrm{sc}}(G) \oplus L^{\mathrm{sg}}(G)$.

Proof. Write every 2×2 block

$$B_{u,v} := \widehat{R}\,\widetilde{L}[u, v]\,\widehat{R} \in \mathbb{R}^{\mathrm{Bool}\,\times\,\mathrm{Bool}}$$

where $\widetilde{L}[u, v]$ is the $\{u\} \times$ Bool-rows, $\{v\} \times$ Bool-columns sub-block of $\widetilde{L}$. A direct Kronecker calculation gives

$$\big(P_{\mathrm{cov}}\,\widetilde{L}\,P_{\mathrm{cov}}\big)\big[(u, i), (v, j)\big] = (\widehat{R}\,\widetilde{L}[u, v]\,\widehat{R})[i, j] = B_{u,v}[i, j]. \tag{3}$$

Split $B_{u,v}$ into four structural cases:

- **(a)** *Diagonal.* If $u = v$, then by Lemma 4.3, $\widetilde{L}[u, u] = \deg(u)\,\mathbb{I}$, whence $\widehat{R}(\deg(u)\,\mathbb{I})\widehat{R} = \deg(u) \cdot 2\mathbb{I}$, so $B_{u,u}[i, j] = 2\deg(u) \cdot \delta_{ij}$.
- **(b)** *Adjacent, non-wrap.* If $\{u, v\} \in E \setminus W$, then $\widetilde{L}[u, v]$ has -1 on the *diagonal* of Bool (sheets match) and 0 off-diagonal: $\widetilde{L}[u, v] = (-1)\mathbb{I}$. So $B_{u,v}[i, j] = -2\delta_{ij}$.
- **(c)** *Adjacent, wrap.* If $\{u, v\} \in E \cap W$, then $\widetilde{L}[u, v]$ has -1 off the diagonal (sheets flip) and 0 on the diagonal: $\widetilde{L}[u, v] = -\sigma_x$. So $\widehat{R}(-\sigma_x)\widehat{R} = -2\,\mathrm{diag}(1, -1)$, i.e. $B_{u,v}[i, j] = -2\delta_{ij}\,(-1)^i$.
- **(d)** *Non-adjacent.* If $\{u, v\} \notin E$, then $\widetilde{L}[u, v] = 0$ and $B_{u,v} = 0$.

Collecting cases and matching against the defining formulas of L^{sc} and L^{sg}:

$$B_{u,v}[i, j] = 2 \cdot \delta_{ij} \cdot \begin{cases} L^{\mathrm{sc}}[u, v] & i = \bot, \\ L^{\mathrm{sg}}[u, v] & i = \top. \end{cases}$$

Because $P_{\mathrm{cov}}^{-1} = \frac{1}{2}P_{\mathrm{cov}}$ (by $P_{\mathrm{cov}}^2 = 2\mathbb{I}$, exactly as in Section 3), reindexing through φ gives the claimed block-diagonal form. □

Corollary 4.5. $\dim_{\mathbb{R}} \ker \widetilde{L} = \dim_{\mathbb{R}} \ker L^{\mathrm{sc}}(G) + \dim_{\mathbb{R}} \ker L^{\mathrm{sg}}(G)$.

Proof. Kernel dimension is invariant under similarity by an invertible matrix and under reindexing by a bijection of the index set; for a block-diagonal matrix it is additive in the blocks. Apply these three standard linear-algebra facts to the similarity of Theorem 4.4. □

Remark 4.6. In the Lean code these three standard facts are the lemmas

`finrank_ker_toLin'_conj`, `finrank_ker_toLin'_reindex`, `finrank_ker_toLin'_fromBlocks_diag`.

They are proved once, over an arbitrary field, in `KernelDimension.lean` and reused for both the block-decomposition Theorem 3.1 and the cover-splitting Corollary 4.5.

5. $\mathbb{Z}/2$ Cohomology and Balanced Components

The wrap predicate $W \subseteq E$ reifies canonically as a $\mathbb{Z}/2$-valued 1-cochain

$$\sigma_W : E \longrightarrow \mathbb{Z}/2, \qquad \sigma_W(e) := \mathbf{1}_{e \in W}.$$

It is automatically a 1-cocycle (there is no 2-skeleton to impose constraints on it). Its cohomology class $[\sigma_W] \in H^1(G; \mathbb{Z}/2)$ is the obstruction to W arising as the coboundary of a vertex $\mathbb{Z}/2$-colouring: $[\sigma_W] = 0$ iff there exists $\varepsilon : V \to \mathrm{Bool}$ with $W = \{\, \{u, v\} \in E : \varepsilon(u) \neq \varepsilon(v) \,\}$. This is the classical notion of *balance* for signed graphs.

Definition 5.1. A connected component $C \in \pi_0(G)$ is *balanced* iff there is a map $\varepsilon : V \to \mathrm{Bool}$ such that for every edge $\{u, v\} \in E$ with $[u]_{\sim_G} = C$,

$$\{u, v\} \in W \iff \varepsilon(u) \neq \varepsilon(v).$$

We write $\pi_0^{\mathrm{bal}}(G) \subseteq \pi_0(G)$ for the subset of balanced components, and $\beta(G) := \#\pi_0^{\mathrm{bal}}(G)$.

Remark 5.2 (Scope conventions for Definition 5.1). Two conventions deserve to be flagged.

(a) **Component-local vs. graph-global.** The definition quantifies only over edges *inside* the component C; the values $\varepsilon(w)$ for w outside C are unconstrained and play no role. For a *connected* graph the present definition coincides with Harary's original graph-global notion of balance ($[\sigma_W] = 0$ in $H^1(G; \mathbb{Z}/2)$); for disconnected graphs they differ, and *every theorem below uses the component-local version.* Throughout the paper $\beta(G)$ denotes the component-local count.

(b) **Isolated vertices are vacuously balanced.** If $C = \{v_0\}$ is a singleton, the edge quantifier is vacuous and the condition holds with any ε. Consequently every isolated vertex counts towards $\beta(G)$; this is load-bearing for Theorem 6.2 below, because an isolated vertex contributes a zero row to L^{sg} and hence $+1$ to the kernel dimension.

The main combinatorial result of this section relates balance to the fibres of the induced map on connected components.

Theorem 5.3. *Let $\pi : \pi_0(\widetilde{G}) \to \pi_0(G)$ be the map induced by the sheet projection $\widetilde{G} \to G$, $(v, b) \mapsto v$. For every $C \in \pi_0(G)$,*

$$\#\pi^{-1}(C) \;=\; \begin{cases} 2 & C \text{ balanced,} \\ 1 & \text{otherwise.} \end{cases}$$

The proof occupies the remainder of this section. Fix $C \in \pi_0(G)$ and choose a representative $v_0 \in V$ with $[v_0]_{\sim_G} = C$. Set the *candidate lifts*

$$D_\perp := [(v_0, \perp)]_{\pi_0(\widetilde{G})}, \qquad D_\top := [(v_0, \top)]_{\pi_0(\widetilde{G})}.$$

Step 1: every component in the fibre is a candidate lift.

Lemma 5.4. *For every $D \in \pi^{-1}(C)$, either $D = D_\perp$ or $D = D_\top$.*

Proof. Since $\pi([(w,b)]) = [w]_{\pi_0(G)}$ factors through the first coordinate, $D = [(w,b)]$ for some $(w,b) \in \widetilde{V}$ with $[w] = C$. By connectedness of C there is a walk $w \to v_0$ in G; lifting edge by edge and setting

$$\text{next sheet}\big(b,\ \{u,u'\}\big) := \begin{cases} \neg b & \{u,u'\} \in W, \\ b & \text{otherwise}, \end{cases}$$

we obtain a walk $(w,b) \to (v_0,b'')$ in $\widetilde{G}$ for some $b'' \in \mathrm{Bool}$ determined by the walk. So $[(w,b)] = [(v_0,b'')]$, and $D \in \{D_\perp, D_\top\}$. □

Step 2: the balanced case.

Lemma 5.5. *If C is balanced, then $D_\perp \neq D_\top$, whence $\#\pi^{-1}(C) = 2$.*

Proof. Let ε witness balance. Define the *XOR* invariant $\chi : \widetilde{V} \to \mathrm{Bool}$ by

$$\chi(v,b) := \varepsilon(v) \oplus b.$$

We claim χ is constant on every connected component of $\widetilde{G}$ whose base image is C. Indeed, for an edge $(u,b) \sim_{\widetilde{G}} (u',b')$ with $[u]_{\sim_G} = C$,

$$\chi(u,b) \oplus \chi(u',b') = \varepsilon(u) \oplus \varepsilon(u') \oplus b \oplus b' = \mathbf{1}_{\{u,u'\}\in W} \oplus \mathbf{1}_{\{u,u'\}\in W} = 0,$$

using the definition of $\sim_{\widetilde{G}}$ and of balance. So χ is a locally constant function on the fibre over C, hence constant on components. But

$$\chi(v_0,\perp) = \varepsilon(v_0) \oplus \perp \neq \varepsilon(v_0) \oplus \top = \chi(v_0,\top),$$

so $(v_0,\perp)$ and $(v_0,\top)$ lie in different components: $D_\perp \neq D_\top$. Together with Lemma 5.4, this forces $\pi^{-1}(C) = \{D_\perp, D_\top\}$, of cardinality 2. □

Step 3: the unbalanced case.

Lemma 5.6. *If C is not balanced, then $D_\perp = D_\top$, whence $\#\pi^{-1}(C) = 1$.*

Proof. Unbalance means: for every candidate $\varepsilon : V \to \mathrm{Bool}$ there is an edge $\{u,u'\} \in E$ with $[u] = C$ on which $\mathbf{1}_{\{u,u'\}\in W} \neq \varepsilon(u) \oplus \varepsilon(u')$. Take the candidate $\varepsilon_0(v) := \mathbf{1}_{(v,\perp)\sim^*_{\widetilde{G}}(v_0,\perp)}$ (i.e. the characteristic function of the $D_\perp$-fibre projected to the base). If $D_\perp \neq D_\top$, then χ constructed from ε_0 as in Lemma 5.5 would witness balance of C, a contradiction. Hence $D_\perp = D_\top$, and $\pi^{-1}(C)$ is a singleton.

The Lean formalisation uses the deck transformation $\delta(v,b) = (v,\neg b)$ in place of the *ad hoc* ε_0 above: if $D_\perp = D_\top$ fails then δ acts freely on $\pi^{-1}(C)$ with every orbit of size 2, forcing $\#\pi^{-1}(C)$ even and the XOR invariant of Lemma 5.5 becomes a well-defined ε witnessing balance of C. Both routes close the same theorem; the argument is recorded as `componentProj_fiber_card` in `L6_Cohomology.lean`. □

Lemmas 5.4–5.6 together prove Theorem 5.3. An immediate corollary is:

Corollary 5.7. $\#\pi_0(\widetilde{G}) = \#\pi_0(G) + \beta(G)$.

Proof. Summing $\#\pi^{-1}(C)$ over $C \in \pi_0(G)$: each balanced component contributes 2, each unbalanced contributes 1. So $\#\pi_0(\widetilde{G}) = 2\beta(G) + (\#\pi_0(G) - \beta(G)) = \#\pi_0(G) + \beta(G)$. □

6. Recognition Theorems

We now reap the kernel-dimension formulas. Throughout, G is a finite simple connection graph.

Theorem 6.1 (Scalar kernel dimension, Mathlib)**.** $\dim_{\mathbb{R}} \ker L^{\mathrm{sc}}(G) = \#\pi_0(G)$.

This is Mathlib's `SimpleGraph.card_ConnectedComponent_eq_rank_ker_lapMatrix` applied to the ordinary graph Laplacian and used as a primitive.

Theorem 6.2. $\dim_{\mathbb{R}} \ker L^{\mathrm{sg}}(G) = \beta(G)$.

Proof. By Corollary 4.5,

$$\dim\ker \widetilde{L} \;=\; \dim\ker L^{\mathrm{sc}}(G) + \dim\ker L^{\mathrm{sg}}(G),$$

where $\widetilde{L} = L^{\mathrm{sc}}(\widetilde{G})$. Applying Theorem 6.1 to $\widetilde{G}$ and to G, and invoking Corollary 5.7,

$$\#\pi_0(G)+\beta(G) = \#\pi_0(\widetilde{G}) = \dim\ker \widetilde{L} = \dim\ker L^{\mathrm{sc}}(G)+\dim\ker L^{\mathrm{sg}}(G) = \#\pi_0(G)+\dim\ker L^{\mathrm{sg}}(G).$$

Subtracting $\#\pi_0(G)$ yields the claim. □

Theorem 6.3. *For* $\epsilon \in \{\mathrm{flat}, \mathrm{Mob}\}$*,*

$$\dim_{\mathbb{R}} \ker L_\epsilon \;=\; \begin{cases} 2\,\#\pi_0(G) & \epsilon = \mathrm{flat}, \\ \#\pi_0(G) + \beta(G) & \epsilon = \mathrm{Mob}. \end{cases}$$

Proof. By Theorem 3.1, L_ϵ is similar (through the reindex φ) to $\mathrm{diag}(L^{\mathrm{sc}}, B_\epsilon)$ with $B_{\mathrm{flat}} = L^{\mathrm{sc}}$, $B_{\mathrm{Mob}} = L^{\mathrm{sg}}$. So

$$\dim\ker L_\epsilon = \dim\ker L^{\mathrm{sc}} + \dim\ker B_\epsilon.$$

Theorems 6.1 and 6.2 give the two diagonal-block kernel dimensions, yielding the claim. □

Theorem 6.4. $\dim\ker L_{\mathrm{Mob}} < \dim\ker L_{\mathrm{flat}}$ *iff* $\beta(G) < \#\pi_0(G)$*, equivalently, iff* not every *component of* G *is balanced.*

Proof. By Theorem 6.3, the strict inequality reduces to $\#\pi_0(G) + \beta(G) < 2\#\pi_0(G)$, i.e. $\beta(G) < \#\pi_0(G)$. The inequality $\beta(G) \le \#\pi_0(G)$ is immediate from the inclusion $\pi_0^{\mathrm{bal}}(G) \subseteq \pi_0(G)$ of Definition 5.1. □

Remark 6.5 (Pre-registered plan vs. correct statement). Our pre-registered plan contained the assertion

$$(\star) \qquad \dim\ker L^{\mathrm{sg}}(G) \;\stackrel{?}{=}\; \#\pi_0(G) - \#\{C \in \pi_0(G) : |W \cap E(C)| \text{ is odd}\}.$$

This is *false* in general. Example: let $G = K_4$ with $W = \{\{1,2\},\{3,4\}\}$, two disjoint wrap edges forming a matching. The whole graph is one component, the total wrap count is 2 (even), so $(\star)$ predicts $\dim\ker L^{\mathrm{sg}} = 1$. However a brute-force eigenvalue computation shows $\dim\ker L^{\mathrm{sg}} = 0$, which is indeed $\beta(K_4) = 0$ from Theorem 6.2 (the two wrap edges together form a non-trivial cohomology class on K_4). Equation $(\star)$ is correct on $G = C_n$ because $H^1(C_n; \mathbb{Z}/2) \cong \mathbb{Z}/2$ is one-dimensional and total wrap-count mod 2 detects the class. It fails for either of two reasons. First, it fails once $\dim H^1(G; \mathbb{Z}/2) > 1$, i.e. once $|E| > |V| - \#\pi_0(G) + 1$ (by the Euler relation $\dim H^1(G; \mathbb{Z}/2) = |E| - |V| + \#\pi_0(G)$). Second, it fails even when $\dim H^1(G; \mathbb{Z}/2) \le 1$ — in particular on trees, where $\dim H^1 = 0$ — whenever a *bridge* of G lies in W: on a tree with a single wrapped edge, $|W| = 1$ is odd so $(\star)$ predicts $\dim\ker L^{\mathrm{sg}} = 0$, but every tree component is balanced (Theorem 10.1) and $\dim\ker L^{\mathrm{sg}} = \#\pi_0(G)$. Precisely, $(\star)$ holds on a component iff that component is unicyclic-or-tree *and* no bridge of it lies in W.

7. The Cycle Spectrum

We close with an explicit diagonalisation in the case $G = C_n$, the n-cycle. Identify $V(C_n) = \mathbb{Z}/n$, with edges $\{i, i+1 \pmod n\}$. Let $\mathbf{e}_k \in \mathbb{C}^n$ be the Fourier basis vector $\mathbf{e}_k[j] = e^{2\pi ikj/n}$ for $k \in \mathbb{Z}/n$. Then the real-valued eigenvectors of any real circulant are $\mathrm{Re}(\mathbf{e}_k)$ and $\mathrm{Im}(\mathbf{e}_k)$ (with the understanding that $\mathrm{Im}(\mathbf{e}_0) \equiv 0$ and, if n is even, $\mathrm{Im}(\mathbf{e}_{n/2}) \equiv 0$, so those two indices contribute only a single real eigenvector each).

Theorem 7.1 (Flat cycle spectrum)**.** *Let $n \geq 3$. The scalar Laplacian $L^{\mathrm{sc}}(C_n)$ has spectrum*

$$\operatorname{Spec} L^{\mathrm{sc}}(C_n) = \big\{\, 2 - 2\cos(2\pi k/n) \;:\; k \in \mathbb{Z}/n \,\big\},$$

with $\mathbf{e}_k$ an eigenvector for the eigenvalue $2 - 2\cos(2\pi k/n)$.

Theorem 7.2 (Möbius cycle spectrum)**.** *Let $n \geq 3$ and $W = \{\{0,1\}\}$ (one wrap edge). Then $L^{\mathrm{sg}}(C_n)$ has spectrum*

$$\operatorname{Spec} L^{\mathrm{sg}}(C_n) = \big\{\, 2 - 2\cos((2k+1)\pi/n) \;:\; k \in \mathbb{Z}/n \,\big\}.$$

Sketch. The scalar Laplacian is the circulant $2\mathbb{I} - (S + S^{-1})$ for the shift $S : v \mapsto v(\cdot + 1)$, so it commutes with the $\mathbb{Z}/n$-action and is diagonalised by the Fourier basis with shift weight $+1$, giving eigenvalues $2 - (e^{2\pi ik/n} + e^{-2\pi ik/n})$. The signed Laplacian with a single wrap edge is *not* circulant before gauging; it commutes with the twisted shift S_{sg} satisfying $S_{\mathrm{sg}}^n = -\mathbb{I}$. The entrywise gauge transformation $v_j \mapsto \alpha^j v_j$ with $\alpha = e^{i\pi/n}$ (relabelled so the branch cut lies on the wrap edge $\{0,1\}$) converts it into the circulant $2\mathbb{I} - (\alpha S + \alpha^{-1} S^{-1})$, giving eigenvalues $2 - (e^{(2k+1)\pi i/n} + e^{-(2k+1)\pi i/n})$. The full Lean proof in `CycleSpectrum.lean` carries out the Fourier algebra without appealing to gauge equivalence, producing each eigenvector explicitly and verifying the eigen-equation by a direct `Finset.sum_eq_single` reduction. □

Remark 7.3. What `CycleSpectrum.lean` formalises is the *per-k eigenvalue identity*: for each $k \in \operatorname{Fin} n$ it produces an explicit nonzero vector v_k with $L^{\mathrm{sc}}(C_n) \cdot v_k = (2 - 2\cos(2\pi k/n))\, v_k$ (`scalarCycle_eigenvalue`, $n \geq 3$) and similarly for the signed Möbius cycle (`signedCycle_eigenvalue`). Assembly of these per-k identities into the full multiset equality $\operatorname{Spec} L^{\mathrm{sc}}(C_n) = \{2 - 2\cos(2\pi k/n)\}$ of Theorems 7.1–7.2 follows by a dimension count, and is not itself a named Lean theorem.

Corollary 7.4. *Let $G = C_n$ with $n \geq 3$ and let $W = \{\{0,1\}\}$ consist of a single wrap edge. Then $\ker L^{\mathrm{sg}}(C_n) = 0$, equivalently $\beta(C_n) = 0$, and hence*

$$\dim\ker L_{\mathrm{Mob}}(C_n, W) = 1 \;<\; 2 = \dim\ker L_{\mathrm{flat}}(C_n)$$

for every $n \geq 3$.

Proof. By Theorem 7.2 the signed eigenvalues are $2 - 2\cos\big((2k+1)\pi/n\big)$ for $k \in \mathbb{Z}/n$. Such an eigenvalue can vanish only if $(2k+1)/n$ is an even integer, which is impossible since $2k+1$ is odd. Hence the signed kernel is zero. The single wrap edge gives odd holonomy around the fundamental cycle, so the unique component is unbalanced, confirming $\beta(C_n) = 0$. The kernel dimensions then follow from Theorem 6.3. □

Remark 7.5. The parity-in-n variant arises for the all-edges-wrap pattern $W = E(C_n)$: the total wrap parity around the fundamental cycle is then $n \bmod 2$, so C_n is balanced exactly when n is even.

8. The Lean Formalisation

The entire development is formalised in Lean 4 against Mathlib v4.11.0. The dependency graph is linear:

$$\texttt{Basic} \to \texttt{KernelDimension} \to \texttt{L5_Cover} \to \cdots \to \texttt{L17_TracesAndLipschitz},$$

with a lateral `CycleSpectrum` file depending on `Basic` and `KernelDimension` (the full file list is tabulated in Appendix A).

Axiomatic footprint. Every named theorem of this paper was checked by the `#print axioms` command to depend only on the three standard Lean axioms `propext`, `Classical.choice`, and `Quot.sound`; none involve `sorryAx` or any user-introduced axiom. In particular:

- Theorem 3.1 is `ConnGraph.laplacian_decomposes`;
- Theorem 4.4 is `ConnGraph.scalarLap_cover_splits`;
- Theorem 5.3 is `ConnGraph.componentProj_fiber_card`;
- Theorem 6.2 is `ConnGraph.signedLaplacian_kernel_dim_general`;
- Theorem 6.3 is `ConnGraph.connectionLaplacian_kernel_dim_general`;
- Theorem 6.4 is `ConnGraph.mobius_kernel_strict_iff_general`;
- Theorems 7.1–7.2 are certified per eigenvalue by `CycleSpectrum.scalarCycle_eigenvalue` and `CycleSpectrum.signedCycle_eigenvalue`; see Remark 7.3 below.

Numerical cross-check. As an independent safety net against the formal statements being merely *provable* but not intended, the accompanying Python script `findings/round2/fuzzer/fuzz.py` enumerates every simple graph on $n \in \{2, 3, 4, 5\}$ vertices up to isomorphism (51 graphs, matching OEIS A000088) and every $2^{|E|}$ wrap subset, totalling 3,456 configurations. For each (G, W) it checks:

- Theorem 4.4 via spectrum equality of $\widetilde{L}$ and $L^{\mathrm{sc}}(G) \oplus L^{\mathrm{sg}}(G)$;
- Corollary 4.5 via SVD-based integer kernel-dimension arithmetic;
- Theorem 5.3 via brute-force $\mathbb{Z}/2$-colouring search and connected-component counting.

All 3,456 configurations agreed with the formal theorem statements; zero disagreements were found.

Summary of effort distribution. The "mathematical" content, in descending order of novelty:

(1) The fibre-cardinality theorem (Theorem 5.3) has no counterpart in Mathlib; it requires the walk-lift construction, XOR-parity invariant, and deck-action freeness argument, all of which are built from scratch against Mathlib's `SimpleGraph.ConnectedComponent` eliminators.

(2) The cover-splitting Theorem 4.4 is an entrywise matrix identity. Mathlib's Kronecker library (`Matrix.mul_kronecker_mul`, etc.) mechanises $\sim 15\%$; the remaining work is a four-case matching on wrap/non-wrap adjacency and a 2x2 block calculation.

(3) The block-decomposition Theorem 3.1 reuses the Kronecker pattern from (2) on the Fin 2 fibre instead of the Bool fibre.

(4) Recognition Theorems 6.2 and 6.3 are a few lines of arithmetic conclusion once (1)–(3) are in place.

Lessons. The formalisation surfaced one genuine bug in our informal plan (the false equation $(\star)$). This is a modest-but-real example of Lean's mathematical utility: the computer cannot be negotiated with when $\beta(G)$ and $\#\pi_0(G) - \#\{\text{odd-wrap components}\}$ fail to agree on K_4 with a matching wrap.

Acknowledgements. This work was carried out as an independent formal-verification project on combinatorial spectral identities, under a pre-registered discovery and adversarial-review discipline.

9. Further Identities: Traces, Bounds, and the Characteristic Polynomial

The block decomposition (Theorem 3.1) and cover splitting (Theorem 4.4) imply a family of exact spectral identities that sit one inference-step away from the recognition theorems. We collect the ones that admit a one-paragraph proof and are stated at the level of generality on which the formalised library already operates. Together they close the "trace-and-inequality gap" left open in earlier versions of this manuscript. All are formalised in

`ConnectionLaplacian/L9_Bounds.lean`, and the multiplicative spectral partition (Theorem 9.6) is formalised in `ConnectionLaplacian/L10_CoverCharpoly.lean`.

Proposition 9.1 (Trace formulas)**.** *For every connection graph $G = (V, E, W)$,*

$$\begin{aligned} \operatorname{tr} L^{\mathrm{sc}}(G) &= 2\,\#E, & \operatorname{tr} L^{\mathrm{sg}}(G) &= 2\,\#E, \\ \operatorname{tr} L_{\mathrm{flat}}(G) &= 4\,\#E, & \operatorname{tr} L_{\mathrm{Mob}}(G) &= 4\,\#E. \end{aligned}$$

Proof. The diagonal entry of L^{sc} at v is $\deg(v)$; irreflexivity of the underlying simple graph kills the wrap off-diagonal contribution at the diagonal, so the diagonal of L^{sg} is also $\deg(v)$; and the diagonals of $L_{\mathrm{flat}}, L_{\mathrm{Mob}}$ are each $\deg(v) \cdot \mathbb{I}_2$, contributing $2\deg(v)$. Summing over v and applying the degree-sum identity $\sum_v \deg(v) = 2\,\#E$ (Mathlib's `SimpleGraph.sum_degrees_eq_twice_ca` proves all four equalities. □

Proposition 9.2 (Quadratic trace and Frobenius identities)**.** *For every connection graph $G = (V, E, W)$,*

$$\begin{aligned} \operatorname{tr}\big(L_{\mathrm{Mob}}(G)^2\big) &= \operatorname{tr}\big(L^{\mathrm{sc}}(G)^2\big) + \operatorname{tr}\big(L^{\mathrm{sg}}(G)^2\big), \\ \|L_{\mathrm{Mob}}(G)\|_F^2 &= \|L^{\mathrm{sc}}(G)\|_F^2 + \|L^{\mathrm{sg}}(G)\|_F^2, \end{aligned}$$

and the cross term obeys the polarisation identity

$$2\,\operatorname{tr}\big(L^{\mathrm{sc}}(G) \cdot L^{\mathrm{sg}}(G)\big) = \operatorname{tr}\big((L^{\mathrm{sc}}(G)+L^{\mathrm{sg}}(G))^2\big) - \operatorname{tr}\big(L^{\mathrm{sc}}(G)^2\big) - \operatorname{tr}\big(L^{\mathrm{sg}}(G)^2\big).$$

Proof. Starting from the Kronecker-conjugation similarity of Theorem 3.1 in its Möbius form, the trace-of-square and Frobenius identities follow from invariance of $\operatorname{tr}(M^2)$ and of $\operatorname{tr}(M^\top M)$ under a same-sided reindex composed with the involution $P = \mathbb{I}_V \otimes_K \widehat{R}$ with $P^{-1} = \frac{1}{2}P$. The cross-term identity is pure algebra: expand $(L^{\mathrm{sc}} + L^{\mathrm{sg}})^2$ and use symmetry of the scalar and signed Laplacians together with the cyclic property $\operatorname{tr}(L^{\mathrm{sc}}L^{\mathrm{sg}}) = \operatorname{tr}(L^{\mathrm{sg}}L^{\mathrm{sc}})$. The Lean names are `trace_sq_laplacian_decomposes`, `frobenius_laplacian_decomposes` and `trace_mul_scalar_signed_` □

Remark 9.3 (Polarisation versus combinatorial form)*.* The cross-term identity above is stated in *polarisation form* and is pure matrix algebra. An equivalent combinatorial form $\operatorname{tr}(L^{\mathrm{sc}} \cdot L^{\mathrm{sg}}) = \operatorname{tr}(L^{\mathrm{sc}2}) - 4\,\#W$, which can be read off the entrywise formula for $L^{\mathrm{sc}} \cdot L^{\mathrm{sg}}$ by summing contributions at diagonal and off-diagonal positions, is confirmed numerically by the R6 Stage-6 fuzzer (zero deviation across 300 random connected graphs at $n = 7$) but is not yet a separately named Lean theorem; a $\leq$ 20-line bridge lemma is recorded as an R7 carry-over.

Proposition 9.4 (Kernel inequalities)**.** *For every connection graph G,*

$$\begin{aligned} \dim\ker L^{\mathrm{sg}}(G) &\leq \dim\ker L^{\mathrm{sc}}(G), \\ \dim\ker L_{\mathrm{flat}}(G) - \dim\ker L_{\mathrm{Mob}}(G) &= \#\pi_0(G) - \beta(G), \end{aligned}$$

and in particular the difference is bounded by $\#\pi_0(G)$.

Proof. By Theorems 6.1, 6.2, 6.3, the LHS of the first line equals $\beta(G)$, the RHS equals $\#\pi_0(G)$, and $\beta(G) \leq \#\pi_0(G)$ because $\pi_0^{\mathrm{bal}}(G) \subseteq \pi_0(G)$. For the second line, $\dim\ker L_{\mathrm{flat}} = 2\,\#\pi_0(G)$ and $\dim\ker L_{\mathrm{Mob}} = \#\pi_0(G) + \beta(G)$, whose difference is $\#\pi_0(G) - \beta(G)$, itself bounded by $\#\pi_0(G)$. □

Lemma 9.5 (Positive semidefiniteness of the signed Laplacian)**.** $L^{\mathrm{sg}}(G) \succeq 0$*. Indeed*

$$\langle x, L^{\mathrm{sg}}(G)\, x\rangle = \sum_{e=\{u,v\}\in E\setminus W} (x_u - x_v)^2 + \sum_{e=\{u,v\}\in W} (x_u + x_v)^2 \qquad \text{for all } x \in \mathbb{R}^V.$$

Proof. Fix $x \in \mathbb{R}^V$. The off-diagonal entry of L^{sg} at (u, v) equals $+1$ when $\{u, v\} \in W$ and -1 when $\{u, v\} \in E \setminus W$; the diagonal is $\deg(u)$. Since L^{sg} is symmetric, each unordered

edge $\{u,v\}$ contributes the pair of entries (u,v) and (v,u) to the quadratic form, yielding a factor of 2:

$$\langle x, L^{\mathrm{sg}}x\rangle \;=\; \sum_{u} \deg(u)\, x_u^2 \;+\; 2\sum_{\{u,v\}\in W} x_u x_v \;-\; 2\sum_{\{u,v\}\in E\setminus W} x_u x_v.$$

Rewriting $\sum_u \deg(u)\, x_u^2 = \sum_{\{u,v\}\in E}\big(x_u^2 + x_v^2\big)$ (each edge contributes to the degree of both endpoints) and splitting the degree sum along $E = (E\setminus W)\sqcup W$:

$$\langle x, L^{\mathrm{sg}}x\rangle \;=\; \sum_{\{u,v\}\in E\setminus W} (x_u^2 + x_v^2 - 2x_u x_v) \;+\; \sum_{\{u,v\}\in W} (x_u^2 + x_v^2 + 2x_u x_v),$$

which is exactly the announced sum of squares. Every summand is ≥ 0, so $\langle x, L^{\mathrm{sg}}x\rangle \geq 0$. □

Theorem 9.6 (Multiplicative spectral partition: cover characteristic polynomial)**.** *Let $\widetilde{L} = L^{\mathrm{sc}}(\widetilde{G})$ be the scalar Laplacian of the orientation double cover of G. Then the characteristic polynomials factor as*

$$\det\big(\lambda\,\mathbb{I}_{\widetilde{V}} - \widetilde{L}\big) \;=\; \det\big(\lambda\,\mathbb{I}_V - L^{\mathrm{sc}}(G)\big)\cdot\det\big(\lambda\,\mathbb{I}_V - L^{\mathrm{sg}}(G)\big).$$

Equivalently, as multisets the spectra satisfy $\mathrm{Spec}(\widetilde{L}) = \mathrm{Spec}(L^{\mathrm{sc}}(G)) \uplus \mathrm{Spec}(L^{\mathrm{sg}}(G))$.

Remark 9.7 (Direct-bundle factorisation). Composing Theorem 9.6 with the similarity of Theorem 3.1 gives the same factorisation directly for L_{Mob} without invoking the cover:

$$\det\big(\lambda\,\mathbb{I} - L_{\mathrm{Mob}}(G)\big) \;=\; \det\big(\lambda\,\mathbb{I} - L^{\mathrm{sc}}(G)\big)\cdot\det\big(\lambda\,\mathbb{I} - L^{\mathrm{sg}}(G)\big),$$

and for every $\alpha\in\mathbb{R}$, $\det(L_{\mathrm{Mob}} + \alpha\mathbb{I}) = \det(L^{\mathrm{sc}} + \alpha\mathbb{I})\cdot\det(L^{\mathrm{sg}} + \alpha\mathbb{I})$. As multisets, $\mathrm{Spec}(L_{\mathrm{Mob}}) = \mathrm{Spec}(L^{\mathrm{sc}})\uplus\mathrm{Spec}(L^{\mathrm{sg}})$. These are Lean theorems `mobius_charpoly_eq_scalar_times_signed`, `shiftedDet_factorises` and `mobius_charpoly_roots_eq_union`.

Proof. By Theorem 4.4, $\widetilde{L}$ is similar (through an invertible Hadamard–Kronecker change of basis composed with the reindex $V\times\mathrm{Bool}\cong V\sqcup V$) to the block matrix $\mathrm{diag}(L^{\mathrm{sc}}(G), L^{\mathrm{sg}}(G))$. Similarity preserves the characteristic polynomial, and the characteristic polynomial of a block-diagonal matrix factors into the product of its diagonal blocks' characteristic polynomials. The multiset equality follows by equating roots with multiplicity. □

Remark 9.8 (Interlacing as a corollary). Theorem 9.6 strictly strengthens the Cauchy interlacing one might expect between the spectra of L^{sg} and $\widetilde{L}$: interlacing would merely bound each eigenvalue of L^{sg} between two eigenvalues of $\widetilde{L}$, whereas Theorem 9.6 pins down the spectrum of $\widetilde{L}$ exactly as the disjoint union of the two smaller spectra. In particular $\det\big(\lambda\mathbb{I} - L_{\mathrm{Mob}}(G)\big) = \det\big(\lambda\mathbb{I} - L_{\mathrm{flat}}(G)\big)$ whenever every component of G is balanced.

Corollary 9.9 (Pointwise multiplicity)**.** *For every $\lambda\in\mathbb{R}$,*

$$m_\lambda(\widetilde{L}) \;=\; m_\lambda(L^{\mathrm{sc}}(G)) + m_\lambda(L^{\mathrm{sg}}(G)),$$

where $m_\lambda(M)$ denotes the multiplicity of λ as an eigenvalue of M.

Proof. Immediate from Theorem 9.6: the multiplicity of λ in a product of polynomials is the sum of multiplicities in the factors. □

10. Structure Theorems for Trees, Forests, and Cycles

We record the "topological" side of the recognition theorems: on graphs where H^1 is understood, the content of Theorem 6.2 becomes explicit.

Theorem 10.1 (Tree balance)**.** *If the underlying simple graph of G is a tree, then every component of G is balanced, for* every *choice of wrap set $W\subseteq E(G)$. Consequently*

$$\beta(G) = \#\pi_0(G), \qquad \dim\ker L^{\mathrm{sg}}(G) = \#\pi_0(G), \qquad \dim\ker L_{\mathrm{Mob}}(G) = 2\,\#\pi_0(G),$$

and the strict inequality of Theorem 6.4 fails.

Proof. Fix a root r of each tree component. For v in that component, let $p_{r\to v}$ be the unique simple $r \to v$ path (Mathlib's `SimpleGraph.IsAcyclic.path_unique`). Define $\varepsilon(v) := \sum_{e\in p_{r\to v}} \mathbf{1}_{e\in W} \pmod 2$, with $\varepsilon(r) = \bot$ by the empty sum. We show $\varepsilon(u) \oplus \varepsilon(v) = \mathbf{1}_{\{u,v\}\in W}$ for every edge $\{u, v\} \in E$ in the component.

Consider the closed walk $w := p_{r\to u} \cdot \{u, v\} \cdot p_{v\to r}$. Because the underlying graph is acyclic, the only closed walks at r are back-tracking compositions of simple edges, so $\sum_{e\in w} \mathbf{1}_{e\in W} \equiv 0 \pmod 2$ (every traversed edge is traversed an even number of times). Splitting the sum into the $p_{r\to u}$-part, the single edge $\{u, v\}$, and the $p_{v\to r}$-part (the latter summing to $\varepsilon(v)$ by symmetry of $\mathbf{1}_{\cdot\in W}$) gives

$$0 \;\equiv\; \varepsilon(u) + \mathbf{1}_{\{u,v\}\in W} + \varepsilon(v) \pmod 2,$$

hence $\varepsilon(u) \oplus \varepsilon(v) = \mathbf{1}_{\{u,v\}\in W}$. Thus ε witnesses balance (Definition 5.1), and every tree component is balanced. The kernel-dimension consequences are immediate from Theorems 6.2 and 6.3. □

Corollary 10.2. *If the underlying graph is a connected tree, then* $\dim\ker L_{\mathrm{Mob}}(G) = 2$ *for every W. In particular paths P_n and stars $K_{1,n}$ have kernel dimension 2 independently of W. The statement generalises to disconnected forests with* $\dim\ker L_{\mathrm{Mob}}(G) = 2\,\#\pi_0(G)$ *by applying the connected case per-component, but this lift is not presently a separately named Lean theorem — only the `IsTree` (connected-acyclic) case is formalised.*

Theorem 10.3 (Walk-sum detector for balance)**.** *Let $C \in \pi_0(G)$. Then C is balanced iff every closed walk $w : v \to v$ with $v \in C$ satisfies $\sum_{e\in w} \mathbf{1}_{e\in W} \equiv 0 \pmod 2$.*

Proof. ($\Rightarrow$) If ε witnesses balance then for each edge $e = \{u, u'\}$ in C, $\varepsilon(u) \oplus \varepsilon(u') = \mathbf{1}_{e\in W}$. Summing along a closed walk $v \to v$ the ε-telescopes, leaving $\sum_e \mathbf{1}_{e\in W} \equiv 0$. ($\Leftarrow$) Pick $v_0 \in C$ and define $\varepsilon(v) := \sum_{e\in w} \mathbf{1}_{e\in W} \pmod 2$ for any walk $w : v_0 \to v$; the closed-walk-even hypothesis makes ε walk-independent. By construction $\varepsilon(u) \oplus \varepsilon(v) \equiv \mathbf{1}_{e\in W}$ on every edge $e = \{u, v\}$ of C, so ε witnesses balance. □

Corollary 10.4 (Cycle even-wrap formula)**.** *If $G = C_n$ is a single cycle of length $n \geq 3$, then*

$$\beta(C_n, W) = \begin{cases} 1 & \#W \equiv 0 \pmod 2, \\ 0 & \#W \equiv 1 \pmod 2, \end{cases} \qquad \dim\ker L^{\mathrm{sg}}(C_n, W) = \begin{cases} 1 & \#W \text{ even}, \\ 0 & \#W \text{ odd}. \end{cases}$$

In particular the pre-registered formula $\dim\ker L^{\mathrm{sg}} = \#\pi_0(G) - \#\{\text{odd-wrap components}\}$ *holds exactly for cycle graphs.*

Proof. The first homology of an n-cycle with $\mathbb{Z}/2$-coefficients is one-dimensional, $H_1(C_n; \mathbb{Z}/2) \cong \mathbb{Z}/2$, generated by the fundamental cycle γ. Every closed walk w at a basepoint decomposes, modulo back-tracking (which contributes edges twice and hence 0 to the $\mathbb{Z}/2$-sum of wrap indicators), into an integer multiple of γ; its wrap-parity therefore equals $[w : \gamma] \cdot \#W \pmod 2$. The walk-sum detector of Theorem 10.3 therefore reduces to the parity of $\#W$ and Theorem 6.2 converts $\beta \in \{0, 1\}$ to $\dim\ker L^{\mathrm{sg}} \in \{0, 1\}$. □

Remark 10.5. The Lean certificate `cycle_isBalanced_iff_even_wrap_weak` (`L14_CycleEw.lean`) names this a *weak* form: it takes an Eulerian closed walk around the cycle and a reverse-direction hypothesis as explicit inputs to the caller. Lifting this to the fully constructive form stated above requires a `fundamentalCycleWalk` helper plus a winding-number argument currently tracked as an R7 carry-over.

11. Counter-claims: What the Formalisation Does *Not* Prove

The extended fuzzer (§8) and the `NEGATOR` pass of our discovery swarm refuted four naive statements that were, at various points, attempted drafts of the identities above. We record them here both for historical honesty and as explicit boundary markers for the theorems that *did* survive.

Example 11.1 (Naive equality of Möbius and scalar kernels is false)**.** A tempting but wrong reformulation of Theorem 6.4 is: "$\dim\ker L_{\mathrm{Mob}} = \dim\ker L_{\mathrm{scalar}}$ iff every component of G is balanced." This fails already on the empty graph on $n = 3$ vertices with $W = \emptyset$: every component is trivially balanced (vacuously so, cf. Remark 5.2), yet $\dim\ker L_{\mathrm{Mob}} = 6$ (a kernel inside $\mathbb{C}^{2V}$) while $\dim\ker L_{\mathrm{scalar}} = 3$ (a kernel inside $\mathbb{C}^{V}$): the two kernels sit in different ambient spaces, so the attempted identity already fails on dimensional grounds. The correct equivalence uses the *flat* Laplacian: $\dim\ker L_{\mathrm{Mob}} = \dim\ker L_{\mathrm{flat}} = 2\,\#\pi_0(G)$ iff every component is balanced, which is Theorem 6.3 combined with Theorem 6.1.

Example 11.2 (A single wrap edge can leave the graph balanced)**.** The implication "G connected and $\#W = 1 \Rightarrow G$ is not balanced" is false. Counterexample: the path P_3 with vertices v_0-v_1-v_2 and the single wrap edge $\{v_0, v_1\}$ (a bridge) is balanced: the assignment $\varepsilon(v_0) = \top,\ \varepsilon(v_1) = \bot,\ \varepsilon(v_2) = \bot$ witnesses balance, since $\varepsilon(v_0) \oplus \varepsilon(v_1) = 1 = \mathbf{1}_{\{v_0,v_1\}\in W}$ and $\varepsilon(v_1) \oplus \varepsilon(v_2) = 0 = \mathbf{1}_{\{v_1,v_2\}\in W}$. The correct form is: a single wrap edge unbalances the graph iff the wrap edge lies on a cycle.

Example 11.3 (Adding a non-wrap edge between components can *decrease* β)**.** The fuzzy-logic sweep (discovery shorthand: claims are labelled `C1`, `C2`, …in the order the swarm proposed them, and $\tau \in [0, 1]$ is the empirical satisfaction rate of a claim over its tested configurations, so $\tau = 0$ refutes and $\tau = 1$ confirms; §8, claim `C2`, $\tau = 0.0000$) refutes the naive monotonicity "adding a non-wrap edge can only preserve or increase β". On three isolated vertices ($W = \emptyset$, so $\beta = 3$), adding the non-wrap edge $\{v_0, v_1\}$ merges two balanced components into one still-balanced component and drops β to 2. The correct law is bidirectional: for a non-wrap edge e joining two distinct components, $\beta(G + e) \in \{\beta(G) - 1, \beta(G)\}$, with -1 when *at least one* endpoint lies in a balanced component and 0 when both endpoints lie in unbalanced components. (In the mixed balanced-plus-unbalanced case, the merged component is unbalanced, so the pre-existing balanced component is lost and β drops by 1.).

Theorem 11.4 (Bridge-deletion monotonicity)**.** *Deleting a non-bridge edge cannot decrease β: for any edge $e \in E(G)$ that is not a bridge of the underlying graph,*

$$\beta(G) \ \le\ \beta(G - e).$$

Proof (formalised). The Lean proof (`L15_BridgeMonotone.lean`, `numBalancedComponents_monotone_remov` constructs an injection from the balanced components of G into the balanced components of $G - e$: deleting a non-bridge edge leaves every component connected, and each balanced component of G yields a balanced component of $G - e$ because its balancing closed walks survive the deletion. A non-wrap hypothesis is accepted by the Lean statement for continuity with the discovery sweep that produced it, but the proof does not use it: the law is bridge-sensitive, not wrap-sensitive. □

Example 11.5 (Non-wrap edge contraction is not kernel-stable)**.** On K_3 with $W = \{\{v_0, v_1\}\}$ the signed Laplacian has kernel dimension 0. Contracting the non-wrap edge $\{v_0, v_2\}$ identifies v_0 and v_2, leaving the single wrap edge $\{v_0, v_1\}$, a tree on 2 vertices: the signed Laplacian now has kernel dimension 1. So non-wrap edge contraction does *not* preserve $\dim\ker L^{\mathrm{sg}}$ in general.

Appendix A. Formalisation index

The principal named theorems, propositions, corollaries, and lemmas of the main text are paired below with corresponding Lean 4 declarations in the `connection_laplacian_lean` sandbox. (A handful of subsidiary lemmas used only in intermediate proofs—e.g. the deck-transformation Lemma 4.2, the cover-degree Lemma 4.3, and the fiber-subset/separation Lemmas 5.4, 5.5, 5.6—are implicitly witnessed inside the proofs of theorems listed below and are not tabulated separately.) The table below pairs them explicitly. All listed declarations are sorry-free and their `#print axioms` output is exactly {`propext`, `Classical.choice`, `Quot.sound`}

— the standard constructive–classical foundation of Mathlib, with no `sorryAx` and no ad-hoc `axiom` declarations. The file paths are relative to `connection_laplacian_lean/ConnectionLaplacian/`.

Paper statement	Lean name	File
Thm. 6.1	`scalarLaplacian_kernel_dim`	`KernelDim.lean`
Thm. 3.1 (direct sum)	`laplacian_decomposes`	`KernelDim.lean`
Thm. 3.1 (ker dim)	`laplacian_kernel_dim_decomposes`	`L8_Recognition.lean`
§4	`scalarLap_cover_splits`	`L5_Cover.lean`
Def. 5.1	`isBalanced`	`L6_Cohomology.lean`
Thm. 5.3	`componentProj_fiber_card`	`L6_Cohomology.lean`
Thm. 6.2	`signedLaplacian_kernel_dim_general`	`L8_Recognition.lean`
Thm. 6.3	`connectionLaplacian_kernel_dim_general`	`L8_Recognition.lean`
Thm. 6.4	`mobius_kernel_strict_iff_general`	`L8_Recognition.lean`
Prop. 9.1 (sc)	`trace_scalarLaplacian`	`L9_Bounds.lean`
Prop. 9.1 (sg)	`trace_signedLaplacianMöbius`	`L9_Bounds.lean`
Prop. 9.4 (mono)	`signed_kernel_le_scalar_kernel`	`L9_Bounds.lean`
Prop. 9.4 (drop $= \beta$)	`kernel_drop_eq_unbalanced`	`L9_Bounds.lean`
Prop. 9.4 (drop $\leq$ $\#\pi_0$)	`kernel_drop_le_numComponents`	`L9_Bounds.lean`
Lem. 9.5	`signedLaplacianMöbius_posSemidef`	`L13_PSD.lean`
Thm. 9.6	`cover_charpoly_eq_scalar_times_mobius`	`L10_CoverCharpoly.lean`
Cor. 9.9	*corollary of `cover_charpoly_eq_scalar_times_mobius`*	
Thm. 10.1	`tree_isBalanced_of_isTree`	`L11_Trees.lean`
(ker dim)	`connectionLaplacian_kernel_dim_tree_mobius`	`L11_Trees.lean`
(gap zero)	`tree_kernel_gap_zero`	`L11_Trees.lean`
Cor. 10.2	*specialisation of the tree ker-dim theorem*	
Thm. 10.3	`isBalanced_iff_closedWalk_wrap_even`	`L12_WalkH1.lean`
Cor. 10.4	`cycle_isBalanced_iff_even_wrap_weak`	`L14_CycleEw.lean`
Ex. 11.3 (R5 refinement, C1 $\tau = 1$)	`numBalancedComponents_monotone_remove_non_bridge`	`L15_BridgeMonotone.lean`
Rem. 9.7 (direct charpoly)	`mobius_charpoly_eq_scalar_times_signed`	`L16_SpectrumUnion.lean`
Rem. 9.7 (spec union)	`mobius_charpoly_roots_eq_union`	`L16_SpectrumUnion.lean`
Rem. 9.7 (shifted det)	`shiftedDet_factorises`	`L17_TracesAndLipschitz.lean`
Prop. 9.1 ($\mathrm{tr}\, L_{\mathrm{Mob}}$)	`trace_laplacian_mobius`	`L17_TracesAndLipschitz.lean`
Prop. 9.2 ($\mathrm{tr}(\cdot^2)$)	`trace_sq_laplacian_decomposes`	`L17_TracesAndLipschitz.lean`
Prop. 9.2 (Frobenius)	`frobenius_laplacian_decomposes`	`L17_TracesAndLipschitz.lean`
Prop. 9.2 (polarisation)	`trace_mul_scalar_signed_eq`	`L17_TracesAndLipschitz.lean`

(File paths: `KernelDim.lean` abbreviates `KernelDimension.lean`.)

Reproducibility. The complete Lean and Python sources live at `https://github.com/jesusvilela/connection_laplacian_lean` (DOI 10.5281/zenodo.19695396). From the project root,

```
elan run lake build
```

rebuilds all 1833 Lean objects (Mathlib v4.11.0 + `ConnectionLaplacian`); the table above can be regenerated by invoking `#print axioms` on each listed name. Continuous fuzz-sweeps now span five rounds of the discovery swarm. The R3/R4 baseline exhausts 176,169 connection-graph configurations ($n \leq 5$ exhaustive; $n \in \{6, 7, 8\}$ wrap-subset / random-graph); R5 extends this to a 456,950-configuration sweep at $n \leq 7$ targeting the cover-charpoly and kernel-dimension identities; R6 Stage-2 `FUZZER-A` (`findings/round6/stage2_fuzzer_A/`) certifies the M6 multiset-union identity at $1.5{\cdot}10^{-15}$ relative precision on 3,727 iso-configurations at $n = 6$ plus a random $n = 7$ sample; R6 Stage-6 `FUZZER-B` (`findings/round6/stage6_fuzzer_B/`) cross-checks the shifted-determinant, trace-square, Frobenius-norm and cross-polarisation identities of L17 at $n = 7$ (300 graphs, 14,011 probes, zero counterexamples). Across all rounds, the only non-integer numerical deviations are 20 out of 24,971 configurations at $n = 8$ that show a 10^{-3}-order relative error in evaluating $\mathrm{charpoly}(\widetilde{L}^{\mathrm{cov}})$ through the

companion matrix (attributable to condition-number growth, bounded by standard float-polynomial perturbation theory; max deviation $7.3 \cdot 10^{-3}$). The underlying Lean identity `cover_charpoly_eq_scalar_times_mobius` is an exact statement over $\mathbb{Z}[\lambda]$ and is unaffected. Fuzzer scripts and JSON outputs live under `findings/round{3,4,5,6}/`.
Counter-claims are also machine-shadowed. Each counter-example in §11 (Examples 11.1, 11.2, 11.3, 11.5) is emitted by the `NEGATOR` / `NEGATOR-FUZZY` passes of the discovery swarm; the generating Python scripts live under `findings/round3/negator/` and `findings/round4/negator_fuzzy/` and can be re-run for any $n \leq 5$ (and $n = 6$ with a wrap-subset cap) to regenerate the counter-witnesses from scratch.

References

[1] F. Harary. *On the notion of balance of a signed graph.* Michigan Math. J. **2** (1953/54), 143–146.
[2] T. Zaslavsky. *Signed graphs.* Discrete Appl. Math. **4** (1982), 47–74.
[3] F. R. K. Chung. *Spectral Graph Theory*, CBMS Regional Conference Series in Mathematics **92**, American Mathematical Society, Providence, 1997.
[4] Y. Bilu and N. Linial. *Lifts, discrepancy and nearly optimal spectral gap.* Combinatorica **26** (2006), 495–519.
[5] The Mathlib Community. *The Lean Mathematical Library.* In: *CPP 2020 Proceedings*, 367–381, 2020.
[6] L. de Moura and S. Ullrich. *The Lean 4 theorem prover and programming language.* In: *CADE 28 Proceedings*, LNCS 12699, 625–635, 2021.

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